

Ductwork Layout Drawing & Specifications

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Project: Lakefront Office Tower — Phase II

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SECTION 23 31 13 — METAL DUCTS

1.0 DUCT CONSTRUCTION STANDARDS

All ductwork shall be constructed and installed per SMACNA HVAC Duct Construction Standards 3rd Edition. Pressure class shall be as scheduled below. Sheet metal gauges per SMACNA Table 1-1.

2.0 MATERIAL SCHEDULE

Duct System	Pressure Class	Material	Min. Gauge	Insulation (ext.)
Supply Main	2-in. SP	G90 Galv. Steel	20 ga.	2-in. duct wrap R-8
Supply Branch	2-in. SP	G90 Galv. Steel	22 ga.	1.5-in. duct wrap R-6
Return Main	1-in. SP	G90 Galv. Steel	22 ga.	None (plenum)
Return Branch	1-in. SP	G90 Galv. Steel	24 ga.	None
Exhaust	1-in. SP	G90 Galv. Steel	24 ga.	None
Kitchen Exhaust	2-in. SP	304 SS, 16 ga.	16 ga.	1-hr fire-wrap
OA / Relief	1-in. SP	G90 Galv. Steel	22 ga.	2-in. R-8 (exterior)

3.0 TYPICAL FLOOR PLAN — DUCT LAYOUT (FLOOR 5)

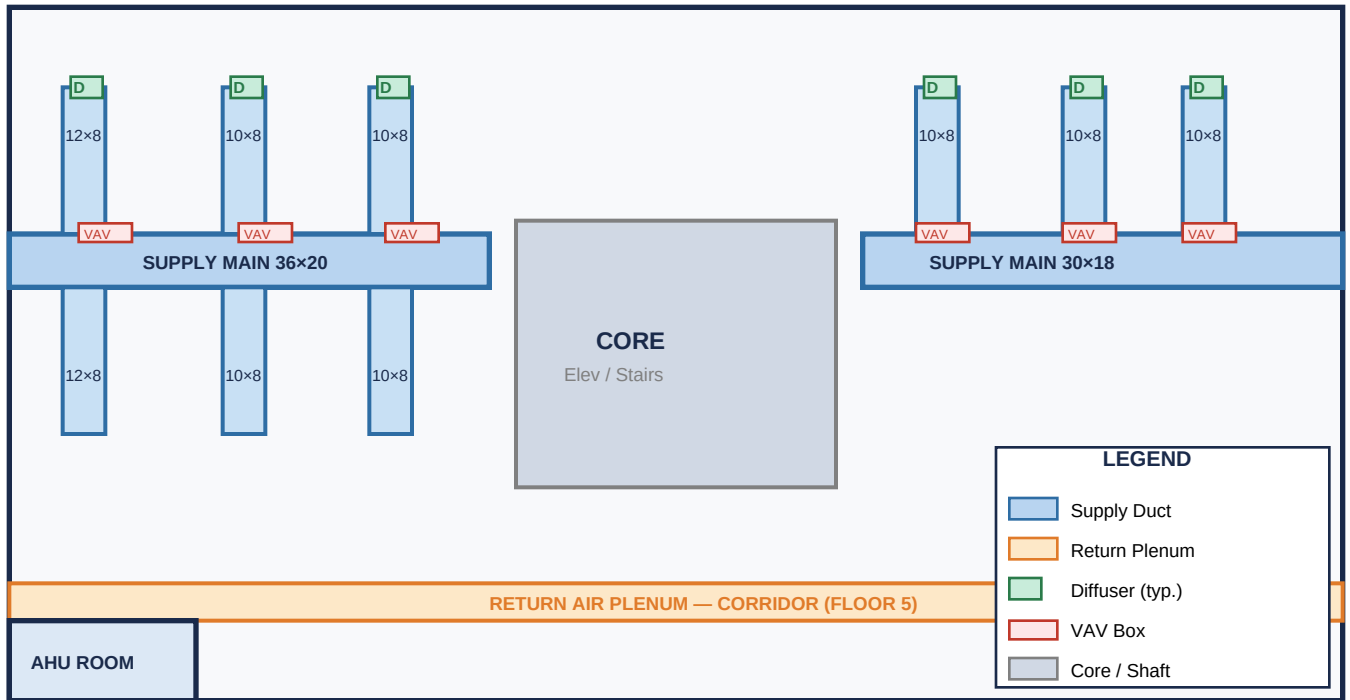


Figure 2 — Typical Floor Duct Layout, Floor 5 (Schematic — Not to Scale)

4.0 VAV BOX SCHEDULE

Tag	Zone	CFM Max	CFM Min	Box Size (in.)	Coil (MBH)	Ctrl
VAV-501	Open Office E	2,400	480	12x12	12	DDC
VAV-502	Open Office W	2,200	440	12x12	12	DDC
VAV-503	Conference 501	800	200	8x8	6	DDC
VAV-504	Conference 502	600	160	8x8	4	DDC
VAV-505	Private Off. A	350	90	6x6	3	DDC
VAV-506	Private Off. B	350	90	6x6	3	DDC
VAV-507	Break Room	500	150	8x6	4	DDC
VAV-508	Reception	400	100	6x6	3	DDC

5.0 INSTALLATION NOTES

- All duct joints to be sealed with UL 181 mastic or foil-faced tape rated for pressure class.
- Flexible duct connections limited to 18 inches maximum length.
- All rectangular duct exceeding 19-in. width: provide intermediate hangers at 4-ft max spacing.
- Provide access doors minimum 12x12 in. at all fire dampers, smoke dampers, and coil valves.
- Leakage testing required for all pressure class 2 duct systems per SMACNA Leakage Class 6.
- Duct insulation vapor barrier facing must be continuous and un-punctured.
- Coordinate duct routing with structural, electrical, and plumbing disciplines before fabrication.

END OF SECTION 23 31 13